



HABIB
UNIVERSITY

Midterm 1 Part A
Electromagnetic Theory
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Q1 [30]: Find the electric field at point P with Cartesian coordinates $(0, 0, h)$ due to a circular disk of radius a residing in the x - y plane, and assume that the surface charge density to vary with r as

$$\rho_s = \rho_0 r^4 \quad (\text{C/m}^2)$$

where ρ_0 is a constant.

Hint: In integral evaluation, one may assume

$$u^2 = r^2 + h^2.$$

Q2 [20]: In a certain region of space, the electric field is given in spherical coordinates as follows:

$$\mathbf{E} = \hat{\mathbf{R}} \frac{20}{R^4} \quad (\text{V/m})$$

Find the electric potential of point A with respect to point B where A is at $+2$ m and B at -4 m, both on the x -axis.

Hint: Since both field points are on x -axis, therefore, one may write

$$\mathbf{E} = \pm \hat{\mathbf{x}} \frac{20}{x^4} \quad (\text{V/m})$$

Q3 [50]: In a certain region of space, the volumetric charge density is given in cylindrical coordinates by the function:

$$\rho_v = 10e^{-r} \quad (\text{C/m}^3)$$

(a) Apply the integral form of Gauss's law $\oint_S \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enclosed}}$ to find $\mathbf{D}$.

(b) Apply the differential form of Gauss's law $\nabla \cdot \mathbf{D} = \rho_v$ to find $\mathbf{D}$.

Hint: In both cases, one may assume $\mathbf{D} = D_r \hat{\mathbf{r}}$.

Q3 Solution:

Method 1: Integral Form of Gauss's Law

Since ρ_v varies as a function of r only, so will $\mathbf{D}$. Hence, we construct a cylinder of radius r and length L , coincident with the z -axis. Symmetry suggests that $\mathbf{D}$ has the functional form $\mathbf{D} = \hat{\mathbf{r}}D$. Hence,

$$\begin{aligned}\oint_S \mathbf{D} \cdot d\mathbf{s} &= Q \\ \int \hat{\mathbf{r}}D \cdot d\mathbf{s} &= D(2\pi rL) \\ Q &= 2\pi L \int_0^r 10e^{-\tilde{r}} \cdot \tilde{r}d\tilde{r} \\ &= 20\pi L [1 - (1+r)e^{-r}] \\ \mathbf{D} = \hat{\mathbf{r}}D &= \hat{\mathbf{r}}\frac{10}{r} [1 - (1+r)e^{-r}]\end{aligned}$$

Method 2: Differential Form of Gauss's Law

$$\nabla \cdot \mathbf{D} = \rho_v, \quad \mathbf{D} = \hat{\mathbf{r}}D_r,$$

with D_r being a function of r .

$$\frac{1}{r} \frac{\partial}{\partial r} (rD_r) = 10e^{-r},$$

We solve it as follows:

$$\begin{aligned}\frac{\partial}{\partial r} (rD_r) &= 10re^{-r}, \\ \int_0^r \frac{\partial}{\partial \tilde{r}} (\tilde{r}D_r) d\tilde{r} &= \int_0^r 10\tilde{r}e^{-\tilde{r}} d\tilde{r} \\ rD_r &= 10 [1 - (1+r)e^{-r}], \\ \mathbf{D} = \hat{\mathbf{r}}D_r &= \hat{\mathbf{r}}\frac{10}{r} [1 - (1+r)e^{-r}]\end{aligned}$$

Q1 Solution:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int_{S'} \hat{\mathbf{R}}' \frac{\rho_s(\mathbf{R}') ds'}{R'^2} = \frac{1}{4\pi\epsilon_0} \int_{S'} \mathbf{R}' \frac{\rho_s(\mathbf{R}') ds'}{R'^3}$$

The position vector of the field point, P , is $\mathbf{R}_P = h \hat{\mathbf{z}}$.

The position vector of the differential element (patch) on the disc is $\mathbf{R}_d = r \hat{\mathbf{r}}$.

The displacement vector is $\mathbf{R}' = \mathbf{R}_P - \mathbf{R}_d = h \hat{\mathbf{z}} - r \hat{\mathbf{r}}$.

Area of the differential patch is $ds = r dr d\phi$.

The magnitude of displacement vector $\mathbf{R}'$ is $R' = \sqrt{h^2 + r^2}$.

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int_{S'} \mathbf{R}' \frac{\rho_s(\mathbf{R}') ds'}{R'^3}$$

(as we're integrating over entire disc, where r is changing from 0 to a ,

so 'no' auxiliary variable, like r' , is needed)

$$= \frac{1}{4\pi\epsilon_0} \int_{r=0}^a \int_{\phi=0}^{2\pi} (h \hat{\mathbf{z}} - r \hat{\mathbf{r}}) \frac{\rho_0 r^4 r dr d\phi}{(r^2 + h^2)^{3/2}}$$

(owing to symmetry, only z -component survives)

$$= \frac{1}{4\pi\epsilon_0} \int_{r=0}^a \int_{\phi=0}^{2\pi} h \hat{\mathbf{z}} \frac{\rho_0 r^4 r dr d\phi}{(r^2 + h^2)^{3/2}}$$

$$= \frac{\rho_0 h \hat{\mathbf{z}}}{2\epsilon_0} \int_{r=0}^a \frac{r^4 r dr}{(r^2 + h^2)^{3/2}}$$

(substitute $u^2 = r^2 + h^2$, we get $u du = r dr$)

$$= \frac{\rho_0 h \hat{\mathbf{z}}}{2\epsilon_0} \int_{u=h}^{\sqrt{a^2+h^2}} \frac{(u^2 - h^2)^2 u du}{u^3} = \frac{\rho_0 h \hat{\mathbf{z}}}{2\epsilon_0} \int_{u=h}^{\sqrt{a^2+h^2}} \frac{(u^4 + h^4 - 2u^2 h^2) u du}{u^3}$$

$$= \frac{\rho_0 h \hat{\mathbf{z}}}{2\epsilon_0} \int_{u=h}^{\sqrt{a^2+h^2}} \left(u^2 + \frac{h^4}{u^2} - 2h^2 \right) du = \frac{\rho_0 h \hat{\mathbf{z}}}{2\epsilon_0} \left(\frac{u^3}{3} - \frac{h^4}{u} - 2h^2 u \right) \Big|_h^{\sqrt{a^2+h^2}}$$

(apply limits yourself to obtain the final expression.)

Q2 Solution:

$$V_{AB} = V_A - V_B = - \int_B^A \mathbf{E} \cdot d\mathbf{l}$$

Along x -direction, $\hat{\mathbf{R}} = \hat{\mathbf{x}}$ and $\mathbf{E} = \hat{\mathbf{x}} \frac{20}{x^4}$ for $x \geq 0$,

and $\hat{\mathbf{R}} = -\hat{\mathbf{x}}$ and $\mathbf{E} = -\hat{\mathbf{x}} \frac{20}{x^4}$ for $x \leq 0$.

This is due to the fact that the field is defined in spherical coordinates where its direction is radially outward away from the origin.

Hence

$$V_{AB} = - \int_{-4}^2 \mathbf{E} \cdot \hat{\mathbf{x}} dx = - \left[\int_{-4}^0 -\hat{\mathbf{x}} \frac{20}{x^4} \cdot \hat{\mathbf{x}} dx + \int_0^2 \hat{\mathbf{x}} \frac{20}{x^4} \cdot \hat{\mathbf{x}} dx \right] = \frac{35}{48} \text{ V}$$